

Section 1 Arbitrage and the Logic of Pricing

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Overview

This section builds arbitrage-free pricing in the simplest setting: one period (“today” and “at maturity”), finitely many states of the world, and no calculus. Four ideas come in order.

- §1.1: replication forces prices.
- §1.2: the formal market model.
- §1.3: no arbitrage holds exactly when positive *state prices* exist. This is the fundamental theorem of asset pricing.
- §1.4: dividing state prices by physical probabilities gives the *stochastic discount factor*. Normalizing them instead gives the *risk-neutral measure*.

Everything later in the course is this same logic on a richer probability space. That includes Black–Scholes, Heston, Monte Carlo and the volatility surface. One external result is imported, the separating hyperplane theorem. Every other statement is proved in full. Proofs are worked in class in the gray blocks and appear in the complete version of these notes.

1.1 The Pricing Paradigm

What kind of pricing does financial engineering do, and what assumption makes it work? It does relative pricing, and the assumption is the absence of arbitrage. Asset pricing comes in two kinds.

- *Equilibrium* pricing asks what an asset is worth in absolute terms. It models investors’ preferences, endowments and beliefs. It clears supply against demand and reads prices off the equilibrium.
- *Relative* pricing takes the prices of some assets as *given* by the market. It then asks what a related asset must cost, so that the two prices are consistent.

Financial engineering is relative pricing. We almost never ask why the stock is worth \$100. We ask what an option on it must be worth, given that it is. That is what “engineering” means here: we build one instrument out of others whose prices we accept as data.

The tool that does the building is *replication*. Suppose a portfolio of traded assets reproduces the payoff of a target instrument, state by state. Then holding the portfolio and holding the instrument give the same future cash flow. Two claims to the same future cash flow must trade at the same price today. This is the *law of one price*. So the price of the target is forced. It equals the cost of building the replicating portfolio. Replication is the pricing idea of this whole course.

The law of one price holds because there is no *arbitrage*. Informally, an arbitrage is a trading strategy that cannot lose and might win. It either costs nothing today and pays a non-negative amount that is strictly positive somewhere, or it pays cash today and owes nothing later. These are the “free lunches” that competitive markets remove. If one existed, traders would scale it without bound. That single assumption is enough to pin down prices, and its full content is the existence of a positive linear pricing rule.

Here is the mechanism in one small market, before we formalize it. Two assets trade over one period, with two states at maturity. A riskless bond pays \$1 in either state and costs \$0.80 today. A stock pays \$2 in state 1 and \$1 in state 2, and costs \$1.40 today. We write payoffs as vectors ordered by state.

Example 1.1. Using only the bond $(1, 1)$ priced at 0.80 and the stock $(2, 1)$ priced at 1.40, find the price of a structured claim X that pays \$4 in state 1 and \$3 in state 2.

We look for a portfolio of a bonds and b stocks whose payoff matches X state by state:

$$a \cdot 1 + b \cdot 2 = 4 \quad (\text{state 1}), \quad a \cdot 1 + b \cdot 1 = 3 \quad (\text{state 2}).$$

Subtracting the second equation from the first gives $b = 1$, and then $a = 3 - b = 2$. So two bonds and one stock replicate X exactly. Its payoff is $2(1, 1) + 1(2, 1) = (4, 3) = X$. By the law of one price, X must cost what that portfolio costs today,

$$\text{price of } X = a(0.80) + b(1.40) = 2(0.80) + 1(1.40) = 1.60 + 1.40 = 3.00.$$

No preferences, no probabilities and no discounting assumption entered. Given the two quoted prices, \$3.00 is the *only* price for X that admits no free lunch. If X traded at \$2.80, you would buy X and short the replicating portfolio. That banks \$0.20 today with nothing owed at maturity. At \$3.20 you would do the reverse.

1.2 The One-Period Market Model

This subsection sets up the formal market: states, assets, portfolios, and the payoffs that trading lets you build. All later notation is fixed here. Time has two dates: $t = 0$ (today) and $t = 1$ (maturity). Uncertainty is described by finitely many *states of the world* $s = 1, 2, \dots, S$. Exactly one of them occurs at $t = 1$. A state is a complete list of everything payoff-relevant that could happen. Finiteness is the simplification that lets us use linear algebra instead of measure theory.

Definition 1.2 (One-period market model).

A one-period market consists of N traded assets. Asset j has a known price p_j today, collected in the price vector $p = (p_1, \dots, p_N)^\top \in \mathbb{R}^N$. Prices are not assumed positive: a forward or a swap can be entered at zero cost, and a contract can carry negative value. Asset j pays D_{sj} units of account at $t = 1$ if state s occurs. The $S \times N$ payoff matrix $D = (D_{sj})$ has one column per asset and one row per state. A portfolio is a vector $\theta = (\theta_1, \dots, \theta_N)^\top \in \mathbb{R}^N$, where θ_j is the number of units of asset j held (negative entries are short positions). Its cost today is the number $p^\top \theta$, and its payoff at $t = 1$ is

the vector $D\theta \in \mathbb{R}^S$, whose s -th entry $(D\theta)_s = \sum_j D_{sj}\theta_j$ is what the portfolio delivers in state s . A payoff vector $x \in \mathbb{R}^S$ is marketed (or attainable) if $x = D\theta$ for some portfolio θ ; the set of marketed payoffs,

$$\mathcal{M} = \{D\theta : \theta \in \mathbb{R}^N\} = \text{range}(D) \subseteq \mathbb{R}^S,$$

is a linear subspace of $\mathbb{R}^S$, the marketed subspace.

Read this economically. The space $\mathbb{R}^S$ holds all conceivable maturity payoffs, one coordinate per state. A point of $\mathbb{R}^S$ is a contingent claim: it promises x_s if state s occurs. Trading lets you manufacture the payoffs in $\mathcal{M}$ and no others. Replication is the statement “ $x \in \mathcal{M}$ ”, and its cost is what you pay to build it. The whole subject lives in the gap between the claims you might want ($\mathbb{R}^S$) and the claims you can build ($\mathcal{M}$). Figure 1 pictures this for $S = 2$.

One feature of $\mathcal{M}$ matters later. It is built from D alone, so it does not depend on the prices p at all. Change every quoted price and $\mathcal{M}$ is unchanged. What the market can *build* and what the market *charges* are separate questions, and only the second one can produce an arbitrage.

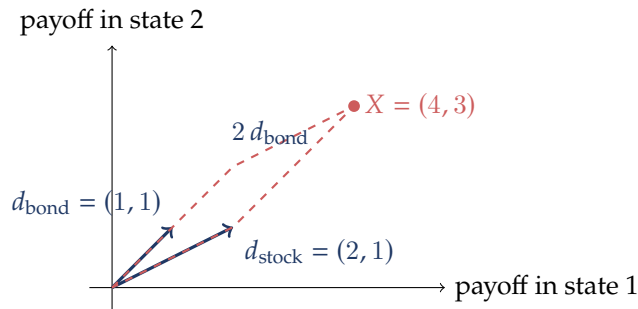


Figure 1: The state-payoff plane for two states. Each traded asset is a vector, and the marketed subspace $\mathcal{M}$ is their span. Two independent assets span the whole plane, so *every* claim is replicable. The market is complete. The claim $X = (4, 3)$ is reached by the parallelogram rule as $2 d_{\text{bond}} + 1 d_{\text{stock}}$, and its price is the cost of that combination.

A market cannot quote its prices freely. If it did, some portfolios would be free lunches. We isolate the two forms this takes.

Definition 1.3 (Arbitrage).

In a one-period market an arbitrage is a portfolio θ of either of the following two types.

Type I:

$p^\top \theta = 0$, $D\theta \geq 0$, and $D\theta \neq 0$ (it costs nothing today, never pays out a negative amount, and pays strictly positive in at least one state: a free lottery ticket).

Type II:

$p^\top \theta < 0$ and $D\theta \geq 0$ (it generates a strictly positive cash inflow today and imposes no liability at maturity: money now, nothing owed later).

Here $D\theta \geq 0$ means every component is non-negative. A market is arbitrage-free if it admits no arbitrage of either type.

The two types are one idea seen from two sides. Type I takes value *tomorrow* for free. Type II takes it *today*. The two are related, but the link needs an extra ingredient. Suppose some traded asset has a positive price and a non-negative, non-zero payoff, as a riskless bond does. Take a

type-II arbitrage returning cash $c > 0$ now and invest that c in that asset. The result costs nothing today and pays a non-negative amount that is positive somewhere, which is a type-I arbitrage. Without such an asset the conversion need not work, so we keep both types on the books. The results below then exclude every free lunch at once.

Notice what the definition does *not* mention: probabilities. Whether a payoff is non-negative in every state is a statement about the states themselves. It says nothing about how likely they are. So arbitrage is robust. It survives any disagreement about probabilities, provided everyone agrees which states are possible.

Some assets carry no independent information. The market already prices their payoff through the others.

Definition 1.4 (Redundant asset).

An asset is redundant if its payoff vector is a linear combination of the other assets' payoff vectors. Equivalently, a portfolio of the remaining assets can replicate the payoff. A market is said to have no redundancy when the columns of D are linearly independent.

A redundant asset is not useless. It may be the very instrument we wish to price. But it adds nothing to the set $\mathcal{M}$ of achievable payoffs. Its payoff is already marketed, so no-arbitrage leaves its price no freedom. That is the content of the next result.

Proposition 1.5 (Law of one price).

In an arbitrage-free market, any two portfolios with the same payoff have the same cost. That is, if $D\theta = D\theta'$ then $p^\top\theta = p^\top\theta'$. In particular the price of a redundant asset equals the cost of any portfolio that replicates it.

With no free lunch, the price of a payoff depends only on the payoff itself. So anything you can build from traded assets must cost exactly what those assets cost to assemble.

Proof.

□

We close the subsection with a worked market. It uses every notion just introduced: marketed payoffs, redundancy, the law of one price, and the detection of an arbitrage.

Example 1.6. A market has three assets over two states: a bond with payoff $(1, 1)$ priced at 0.80; a stock with payoff $(2, 1)$ priced at 1.40; and a structured note with payoff $(4, 3)$ priced

at 2.80. Is the market arbitrage-free? If not, construct an explicit arbitrage.

The bond and stock are the market of Example 1.1. The note's payoff (4, 3) is redundant. We found there that $(4, 3) = 2(1, 1) + 1(2, 1)$, so two bonds and one stock replicate the note, at cost $2(0.80) + 1(1.40) = 3.00$. By the law of one price a no-arbitrage market must price the note at \$3.00. It is quoted at \$2.80, so the market is *not* arbitrage-free. To harvest the mispricing, buy the cheap note and short its expensive replicant. The resulting portfolio is

$$\theta = \left(\underbrace{-2}_{\text{bond}}, \underbrace{-1}_{\text{stock}}, \underbrace{+1}_{\text{note}} \right),$$

that is, short two bonds, short one stock, buy one note. Its cost today is

$$p^\top \theta = -2(0.80) - 1(1.40) + 1(2.80) = -1.60 - 1.40 + 2.80 = -0.20 < 0,$$

a cash inflow of \$0.20, and its payoff at maturity is

$$D\theta = -2(1, 1) - 1(2, 1) + 1(4, 3) = -(4, 3) + (4, 3) = (0, 0) \geq 0.$$

Money today, nothing owed later: a type-II arbitrage of \$0.20 per unit, scalable without limit. The mispricing must disappear, which forces the note back to \$3.00.

1.3 State Prices and the Fundamental Theorem

This subsection answers the central question of the section. When is a market free of arbitrage, and what does that guarantee about prices? The answer is the fundamental theorem of asset pricing. We reach it through state prices.

The law of one price says consistent prices are a linear function of payoffs. We now identify that function. For each state s , imagine an idealised security that pays exactly \$1 if state s occurs and \$0 otherwise. This is an *Arrow security*. If it trades, call its price ψ_s . Any payoff $x = (x_1, \dots, x_S)$ is a bundle of Arrow securities, x_s units of the s -th. So its price must be $\sum_s \psi_s x_s$. This is the pricing rule we are after, and the numbers ψ_s are its coordinates.

Definition 1.7 (State-price vector).

Write $\psi = (\psi_1, \dots, \psi_S)^\top$ for a vector of candidate prices, one per state. We call ψ a *state price vector* if it is strictly positive, $\psi \gg 0$, and reproduces every traded asset's price as the sum of that asset's payoffs weighted by the ψ_s :

$$p_j = \sum_{s=1}^S \psi_s D_{sj} \quad \text{for every asset } j, \quad \text{equivalently} \quad p = D^\top \psi.$$

The number ψ_s is what this pricing rule charges for one unit of account delivered if and only if state s occurs. Read it as an *implicit state price*: a value assigned by a positive linear rule that reproduces every observed asset price. Whether it is also a market price depends on the market. If the Arrow security for state s is attainable, then ψ_s is its unique no-arbitrage price. If it is not attainable, Theorem 1.11

will show that several admissible ψ exist, and they can assign it different values.

Two economic remarks fix the meaning. First, positivity $\psi_s > 0$ is not decoration. A claim that pays a non-negative amount everywhere, and a positive amount in state s , cannot be free. So the price of state- s delivery must be strictly positive. Second, ψ discounts *and* weights at the same time. The rule values a certain dollar at $t = 1$ at $\sum_s \psi_s$ today. Suppose the riskless payoff $(1, \dots, 1)$ is traded, or can be replicated. Then its cost is fixed by the law of one price, and

$$\sum_s \psi_s = \frac{1}{1+r},$$

where r is the riskless rate. Without that assumption the sum is still well defined for each admissible ψ , but it need not be the same for all of them, and there is no single riskless rate to read off. How the total splits across states shows how the market values a dollar by *where* it arrives. Section 1.4 separates these two roles.

Before the main theorem, note that a state-price vector is already hiding inside the law of one price. The law of one price alone, with no mention of arbitrage, forces prices to be linear in payoffs. That is exactly what a vector ψ delivers.

Proposition 1.8 (Law of one price and linear pricing).

The law of one price holds in a market if and only if there is some vector $\psi \in \mathbb{R}^S$, with signs unrestricted, such that $p = D^\top \psi$.

So the law of one price buys a linear pricing rule, but nothing more. It does not make that rule positive.

Proof.

□

Proposition 1.5 said that no-arbitrage implies the law of one price. The reverse is false, and a two-asset market shows it. Let a bond $(1, 1)$ be priced at 1.00 and a stock $(2, 1)$ at 0.50. The two payoff vectors are independent, so no two different portfolios share a payoff and the law of

one price holds automatically. Solving $p = D^\top \psi$ gives $\psi = (-0.5, 1.5)$, which is not positive. And the market does admit a free lunch: short one bond and buy one stock. That portfolio costs $-1.00 + 0.50 = -0.50$ and pays $-(1, 1) + (2, 1) = (1, 0) \geq 0$, which is a type-II arbitrage. So the law of one price can hold while no-arbitrage fails.

The next theorem supplies what is missing, namely positivity. Its hard half rests on one standard result from convex analysis, used here and nowhere else in this course. We state it precisely and import it.

Separating hyperplane theorem. *Let $M \subseteq \mathbb{R}^n$ be a linear subspace and let $K \subseteq \mathbb{R}^n$ be a non-empty, compact, convex set with $M \cap K = \emptyset$. Then there exists a vector $\lambda \in \mathbb{R}^n$ such that*

$$\lambda^\top z = 0 \text{ for all } z \in M, \quad \text{and} \quad \lambda^\top k > 0 \text{ for all } k \in K.$$

Geometrically, a subspace M that never meets a compact convex set K can always be separated from it by a plane through the origin, with all of K strictly on the positive side. The normal λ will turn out to be the state prices, as Figure 2 shows. The result is standard, and proofs go via the closest-point projection onto a closed convex set. See Duffie, *Dynamic Asset Pricing Theory*, 3rd ed., and Björk, *Arbitrage Theory in Continuous Time*, 3rd ed.

Theorem 1.9 (Fundamental theorem of asset pricing, one period).

A one-period market admits no arbitrage if and only if there is a strictly positive state-price vector ψ with which every price satisfies $p = D^\top \psi$.

So a market has no free lunch exactly when a consistent set of positive prices for state-by-state \$1 payoffs reproduces every quoted asset price.

Proof.

□

The theorem converts a question about *strategies* into a question about a *single object*. Instead of asking whether a free lunch exists, we ask whether a positive ψ solves $p = D^\top \psi$. That is linear algebra, which we can always carry out. Whether ψ is unique is a separate matter. It is exactly the question of whether the market can build every payoff.

Definition 1.10 (Complete market).

A market is complete if every payoff is marketed, i.e. $\mathcal{M} = \text{range}(D) = \mathbb{R}^S$; equivalently, the payoff matrix D has rank S (its columns span $\mathbb{R}^S$). Otherwise the market is incomplete.

Completeness says every contingent claim can be replicated. Every risk can be hedged perfectly and every derivative manufactured exactly. It requires at least as many independent assets as states, $\text{rank}(D) = S$, so in particular $N \geq S$. Incompleteness means some claims lie outside $\mathcal{M}$. They cannot be built, and, as we will see, no-arbitrage alone does not price them.

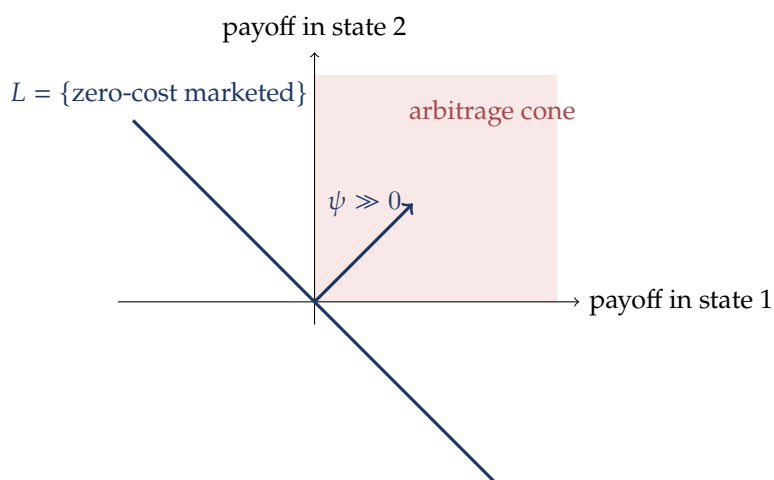


Figure 2: The fundamental theorem, geometrically. Zero-cost marketed payoffs form a line L through the origin, which is a subspace. The arbitrage cone holds the payoffs that are non-negative everywhere and nonzero. No-arbitrage means L meets that cone only at the origin. A separating hyperplane then exists, and its strictly positive normal ψ is precisely the state-price vector. Since $\psi \gg 0$, it prices every non-negative payoff positively.

Theorem 1.11 (Completeness and uniqueness of state prices).

In an arbitrage-free market, the market is complete if and only if the state-price vector is unique.

If the market can build every payoff, the state prices are pinned down to a single vector. If some payoff cannot be built, many different state-price vectors price the traded assets equally well.

Proof.

□

The three market conditions of this section now line up as three statements about the same object, the state-price vector.

Market condition	Equivalent statement about ψ
Law of one price	some $\psi \in \mathbb{R}^S$ solves $p = D^\top \psi$
No arbitrage	some $\psi \gg 0$ solves $p = D^\top \psi$
No arbitrage and completeness	exactly one ψ solves $p = D^\top \psi$

Read the table downwards and each row adds one word. The law of one price gives a pricing rule. No-arbitrage makes that rule *positive*. Completeness makes it *unique*. The three conditions are not equivalent to each other: no-arbitrage implies the law of one price and not the reverse, while completeness neither implies nor is implied by either of the other two. The incomplete market of Example 1.12(b) below is arbitrage-free, and the two-asset market above is complete but admits a free lunch.

The pricing consequence is what matters for the rest of the course. In a complete market every claim has one and only one no-arbitrage price, the cost of its replicating portfolio. In an incomplete market a claim outside $\mathcal{M}$ has a whole *interval* of prices, one for each admissible ψ . We now compute such an interval explicitly.

Example 1.12. (a) In the market with bond $(1, 1)$ at 0.80 and stock $(160, 60)$ at 100 (two assets, two states), find the state-price vector. (b) Now suppose there are three states and the same two assets, with the stock paying $(160, 100, 60)$ and still priced at 100, the bond paying $(1, 1, 1)$ at 0.80. Find *all* state-price vectors and describe the set geometrically.

1.4 The Stochastic Discount Factor and Risk-Neutral Probabilities

This subsection repackages the state-price vector into the two tools used throughout the rest of the course. Both are the same ψ written in different coordinates. Nothing new is assumed. The state space is still finite, so every probability here is just a list of S positive numbers.

Assume throughout that the market is arbitrage-free with state-price vector $\psi \gg 0$. Assume also that a riskless bond is traded, meaning an asset paying \$1 in every state. Its price is $\sum_s \psi_s$,

so the gross riskless return is

$$R = 1 + r = \frac{1}{\sum_s \psi_s}, \quad \text{i.e.} \quad \sum_s \psi_s = \frac{1}{1 + r}.$$

The total mass of the state prices is today's value of a sure dollar.

Now start from a puzzle. Take the complete market of Example 1.12(a): a bond (1, 1) priced at 0.80 and a stock (160, 60) priced at 100, with state prices $\psi = (0.52, 0.28)$ and riskless rate $r = 25\%$. Suppose the true, or *physical*, probabilities are $\pi = (0.8, 0.2)$, so the good state is the likely one.

The first instinct is to price the stock as a discounted expected payoff. The expected payoff is

$$0.8(160) + 0.2(60) = 128 + 12 = 140,$$

and discounting at the riskless rate gives $140/1.25 = 112$. But the stock trades at 100. The instinct is wrong by 12.

One repair is to discount the stock at a higher rate of its own. That rate is $140/100 = 1.40$, a return of 40% rather than 25%. It works for the stock, but it is useless. The bond needs $1/0.80 = 1.25$, and every other asset would need a rate of its own. A rule that changes with the asset is not a pricing rule.

So the correction cannot be a single number. It has to depend on the state. Look for two numbers m_1 and m_2 that reweight the payoff *inside* the expectation, so that one formula prices both traded assets:

$$\pi(X) = \pi_1 m_1 X_1 + \pi_2 m_2 X_2.$$

Writing this out for the bond and for the stock gives two equations,

$$0.8 m_1 + 0.2 m_2 = 0.80, \quad 0.8 m_1(160) + 0.2 m_2(60) = 100.$$

Put $u = 0.8 m_1$ and $v = 0.2 m_2$. The equations become $u + v = 0.80$ and $160u + 60v = 100$. That is exactly the state-price system solved in Example 1.12(a), so $u = 0.52$ and $v = 0.28$, giving

$$m_1 = \frac{0.52}{0.8} = 0.65, \quad m_2 = \frac{0.28}{0.2} = 1.40.$$

The reweighting factor in each state is the state price divided by the physical probability of that state. That object has a name.

Definition 1.13 (Stochastic discount factor).

Given physical probabilities $\pi \gg 0$, the vector $m = (m_1, \dots, m_S)$ defined componentwise by

$$m_s = \frac{\psi_s}{\pi_s}, \quad s = 1, \dots, S,$$

is called the stochastic discount factor, or pricing kernel.

The name is earned by writing the pricing rule as a physical expectation. Since $\psi_s = m_s \pi_s$,

$$\pi(X) = \sum_s \psi_s X_s = \sum_s \pi_s m_s X_s = \mathbb{E}[mX],$$

where $\mathbb{E}$ denotes expectation under the physical probabilities π . Every price is the physical expectation of payoff multiplied by m . So m_s is the state-dependent rate at which future dollars in state s are discounted back to today. Set X to the riskless payoff $(1, \dots, 1)$. This recovers $\mathbb{E}[m] = \sum_s \psi_s = 1/(1+r)$. On average the kernel discounts at the riskless rate. Its variation *across* states is what prices risk.

In the market above we found $m_2 = 1.40$ against $m_1 = 0.65$, so the kernel is more than twice as large in the low-payoff state. Be careful about what that observation is. No-arbitrage delivers a *positive* m and nothing more. It does not say which state is the bad one, and it does not force m to be large there. Here $m_2 > m_1$ is simply what these particular prices imply.

The familiar reading needs economic structure on top. In a consumption-based or representative-agent model, m is the investor's marginal rate of substitution, so m is high in bad states, where marginal utility is high. Under that reading a claim paying off in bad states is expensive, and one paying off in good states is cheap, relative to their physical expected payoffs. That model is where the intuition comes from, not from Theorem 1.9.

The kernel is also the bridge to the empirical asset-pricing literature of the final weeks. Written in returns, $\pi(X) = \mathbb{E}[mX]$ becomes the generic SDF asset-pricing equation $1 = \mathbb{E}[mR_i]$. Identify m with the representative investor's intertemporal marginal rate of substitution and it becomes the consumption-based asset-pricing model. Exercise 1.5 takes the first step.

There is a second way to repackage ψ , and it is the one the rest of the course uses most often. Instead of dividing each state price by a physical probability, divide it by the total $\sum_u \psi_u$. Since $\psi \gg 0$ and that total is $1/(1+r) > 0$, the result is a set of positive numbers summing to one. That is a probability distribution.

Definition 1.14 (Risk-neutral measure).

The risk-neutral measure Q assigns to state s the probability

$$q_s = \frac{\psi_s}{\sum_u \psi_u} = (1+r) \psi_s, \quad s = 1, \dots, S.$$

By construction $q_s > 0$ and $\sum_s q_s = 1$, so Q is a genuine probability distribution on the states.

Substitute $\psi_s = q_s/(1+r)$ into the state-price pricing rule. It becomes an expectation. For a claim with payoff $X = (X_1, \dots, X_S)$,

$$\pi(X) = \sum_s \psi_s X_s = \sum_s \frac{q_s}{1+r} X_s = \frac{1}{1+r} \sum_s q_s X_s = \frac{\mathbb{E}^Q[X]}{1+r}.$$

This is the risk-neutral pricing formula. We *derived* it rather than decreed it. A payoff is valued by its expectation under Q , discounted at the riskless rate.

Read that statement carefully, because Q was built from ψ , and Theorem 1.11 says ψ is unique only when the market is complete. So there is one Q per admissible ψ , and in an incomplete market there are many. Two things follow.

First, if X is *marketed*, say $X = D\theta$, then $\pi(X) = \psi^\top D\theta = (D^\top \psi)^\top \theta = p^\top \theta$ for *every* admissible ψ . The formula returns the cost of the replicating portfolio, and every admissible Q agrees on it. Second, if X is not marketed, different admissible Q generally give different numbers, and

no-arbitrage alone does not choose between them.

Market	Risk-neutral measure	What a claim gets
Complete	exactly one Q	one price, for every claim
Incomplete	many admissible Q	one price if marketed, otherwise an interval

This is the payoff of Section 1.3. Completeness is what turns the pricing rule from a range into a number, and Example 1.12(b) and the exercises compute such a range explicitly.

The name is exact. The market prices *as if* it were risk-neutral, provided we take expectations under Q and not under the real probabilities. The reweighting from π to Q absorbs the risk compensation, so that no risk premium is left to add by hand. Which states gain weight is decided by the traded prices, not by no-arbitrage. In the usual equilibrium reading it is the states investors fear that gain weight, for the same reason the kernel is high there.

The two repackagings do the same job in different ways. The kernel keeps the real probabilities and corrects the payoff. The measure Q leaves the payoff alone and corrects the probabilities.

Notice that the physical probabilities π_s have now dropped out entirely. This is not a paradox. In a complete market the price of X is the cost of its replicating portfolio, and that cost is fixed by the *current prices* of the hedging assets. Those prices already contain the market's views and risk attitudes. So two investors who disagree about how likely state 1 is still agree on the price of every replicable claim. Physical probabilities govern how wealth actually evolves. They are simply not what *relative* prices depend on.

The three objects ψ , m and Q are one thing in three coordinate systems, as Figure 3 shows. Given the riskless rate and π , each is recovered from any other: $\psi_s = m_s \pi_s = q_s / (1+r)$. The kernel and the measure are two readings of the same reweighting. From $q_s = (1+r) \psi_s = (1+r) m_s \pi_s$ we get $q_s / \pi_s = (1+r) m_s = m_s / \mathbb{E}[m]$. So the kernel, rescaled to average one, is exactly the ratio of risk-neutral to physical probabilities.

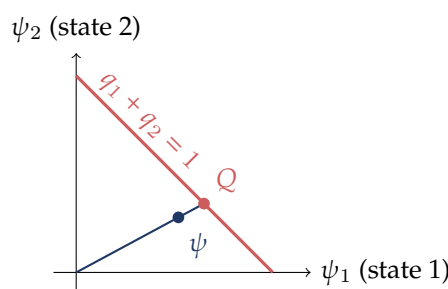


Figure 3: State prices, risk-neutral measure and pricing kernel as one object, drawn for the market of Example 1.12(a). The state-price vector $\psi = (0.52, 0.28)$ is a point in the positive orthant. Scaling it along its ray onto the unit simplex $q_1 + q_2 = 1$ gives the risk-neutral probabilities $Q = 1.25 \psi = (0.65, 0.35)$. The scaling factor is the gross riskless return $1 + r = 1.25$. Dividing ψ componentwise by the physical probabilities π instead gives the stochastic discount factor m . The ray carries all the pricing information, and Q and m are two ways of reading its position.

Example 1.15. Stay in the market of Example 1.12(a): bond $(1, 1)$ at 0.80, stock $(160, 60)$ at 100, state prices $\psi = (0.52, 0.28)$, and physical probabilities $\pi = (0.8, 0.2)$, with kernel $m =$

$(0.65, 1.40)$ as derived above. (a) Find the riskless rate r and verify $\mathbb{E}[m] = 1/(1+r)$. (b) Price the call on the stock struck at $K = 100$, whose payoff is $(\max(160 - 100, 0), \max(60 - 100, 0)) = (60, 0)$, using the kernel, and check the answer by explicit replication. (c) Find the risk-neutral probabilities Q and price the call again under Q . (d) Show the call price does not change if the physical probabilities are changed to $\pi' = (0.5, 0.5)$.

Exercises

Exercise 1.1. A one-period market over two states has a bond paying $(1, 1)$ priced at 0.80 and a stock paying $(100, 50)$ priced at 65. Find the state-price vector ψ , the riskless rate r , and the risk-neutral measure Q . Is the market complete?

Exercise 1.2. A market over two states has three assets: a bond $(1, 1)$ at 0.80, a stock $(100, 50)$ at 65, and a levered note $(150, 100)$ at 102. Show the market admits arbitrage and construct an explicit arbitrage portfolio. State which type it is and the profit per unit.

Exercise 1.3. An incomplete market has three states and two assets: a bond $(1, 1, 1)$ at 0.80 and a stock $(150, 100, 50)$ at 90. (a) Characterize the set of all state-price vectors. (b) A call on the stock struck at $K = 100$ pays $(50, 0, 0)$. Find the interval of its no-arbitrage prices.

Exercise 1.4. Suppose the law of one price fails. That is, two portfolios θ, θ' have the same payoff $D\theta = D\theta'$ but different costs $p^\top \theta \neq p^\top \theta'$. Prove directly that the market admits a type-II arbitrage.

Exercise 1.5. Let a market be arbitrage-free with physical probabilities $\pi \gg 0$ and stochastic discount factor m , so that every asset with price p_i and payoff X_i satisfies $p_i = \mathbb{E}[mX_i]$. Write the gross return of asset i as $R_i = X_i/p_i$ and let $R_f = 1 + r$ be the gross riskless return. Derive the covariance representation of the risk premium,

$$\mathbb{E}[R_i] - R_f = -R_f \text{Cov}(m, R_i),$$

and verify it numerically in the market of Example 1.15 (with $\pi = (0.8, 0.2)$, $m = (0.65, 1.40)$) using the stock.

Exercise 1.6. A trinomial one-period market has a stock priced at 100 paying $(130, 100, 80)$ across three states and a bond $(1, 1, 1)$ priced at 0.80 (so $r = 25\%$). Consider a call struck at $K = 90$, with payoff $(40, 10, 0)$. Show first that the call is not replicable, then find its no-arbitrage price interval, and exhibit the super-replicating portfolio attaining the upper bound and the sub-replicating portfolio attaining the lower bound.